Numerov Method for Bound States in the One-Dimensional Schrodinger Equation

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1 Introduction

This project studies the one-dimensional time-independent Schrödinger equation using the Numerov method combined with parity-aware shooting procedures. The goals are to validate the method on systems with known analytic solutions, demonstrate numerical convergence and accuracy, and then apply it to more interesting bound-state potentials such as a double well and a finite square well. Extensions to scattering problems are also explored as a secondary component.

This problem is a boundary-value eigenvalue problem rather than a standard initial-value problem: only special energies produce wavefunctions satisfying the required boundary behavior. The Numerov method is well suited to this task because the time-independent Schrödinger equation has the second-order form $^{00} = q(x)$, with no first-derivative term. The project therefore focuses not only on obtaining eigenvalues, but also on checking that the numerical results are physically meaningful through analytic benchmarks, convergence tests, parity and node structure, and probability conservation in the scattering extension.

The full code was written in Python and is available in this repository. Explanations on the code structure, usage instructions, and details of the implementation are provided in the repository with the README.md, README.md, and comments in the code. The implementation is modular, with separate components for the Numerov integrator, shooting method, root finding, analysis utilities, and an automated validation suite. This structure allows for independent validation of each numerical component and makes the overall workflow easier to understand.

2 Computational approach

2.1 The Schrodinger equation and Numerov method

In dimensionless units, the Schrödinger equation is written as:

$$^{00}(x) = 2\,[V(x) \quad E]\,(x) \tag{1}$$

The wavefunction is integrated with the Numerov scheme on a uniform grid. For the symmetric bound-state potentials studied here, $V(x) = V(x)$, so the eigenstates can be chosen to have definite parity: even states satisfy $(x) = (x)$ and odd states satisfy $(x) = (x)$. That symmetry implies exact conditions at the origin, $^{0}(0) = 0$ for even states and $(0) = 0$ for odd states, so the solver can integrate only on the half-domain $[0; x_{max}]$ and reconstruct the other half by reflection. Eigenvalues are found by scanning for sign changes in a scalar mismatch function and then refining the resulting brackets with robust root finding; in the outward Numerov solver this is bisection followed by a few safeguarded secant-style polishing steps, while the inward-decay and RK4 solvers use straight bisection. The correctness of the numerical approach is verified through comparison with analytical solutions where available, through systematic convergence studies, and through direct automated tests of the implementation.

The overall numerical workflow is implemented in a modular way in the accompanying Python code. The core components include a Numerov integrator for propagating the wavefunction, a shooting module that enforces parity-based boundary conditions and constructs the mismatch function, and analysis utilities for convergence studies and parameter sweeps. This separation of concerns allows each numerical component to be validated independently and makes the implementation easier to interpret [1].

2.2 Shooting and root nding

The boundary-value eigenproblem is reformulated as a root-finding problem for a mismatch function $M(E)$. For a given trial energy E, the Schrödinger equation is integrated and a boundary mismatch is evaluated. Because the potential is symmetric, the half-domain solution is enough:

once parity is chosen, the left half of the wavefunction is fixed by reflection and the unknown energy is the only quantity left to determine. The code uses two related formulations. For the infinite square well, finite square well, and quartic double well, the half-domain problem is shot outward from $x = 0$ with parity-consistent startup values and the raw mismatch is the outer boundary value $M(E) = \ _E(x_{max})$. This outward formulation is the more general one in the project, because the startup comes from the exact parity conditions at the symmetry point and therefore applies to any symmetric potential once the half-domain reduction is valid. For the harmonic oscillator and the RK4 comparison, the code instead shoots inward from x_{max} through the decaying tail and enforces parity at the origin:

$$\text{even states:} \quad M(E) = \ \frac{\partial}{\partial E}(0); \qquad \text{odd states:} \quad M(E) = \ _E(0) \tag{2}$$

The inward formulation is more specialized: it assumes that the chosen truncation point lies in a forbidden tail where the physical solution is locally exponential, so it is used here for smooth confining infinite-domain problems rather than as a universal replacement for the outward method. The two formulations also differ slightly in how the brackets are polished once a sign change has been found. The outward solver bisects the wall-mismatch bracket and then performs a few safeguarded secant-style updates inside the final sign-changing interval, which improves the final raw boundary residual without giving up bracketing robustness. The inward and RK4 solvers use straight bisection on the origin-parity mismatch instead, which keeps that specialized formulation simpler and fully bracketed. Eigenvalues correspond to the roots of $M(E) = 0$, which are located by bracketing sign changes and then refined numerically. To make the procedure transparent, the report now uses two complementary diagnostics: a global raw-mismatch scan over the full energy interval, where only the final root estimates are marked, and per-root zoom plots that restrict the horizontal axis to one initial bracket and show the full bisection history. Zero crossings identify the allowed energies.

In the implementation, the inward even-state mismatch uses a fourth-order backward finite-difference stencil to approximate $\ ^{\prime}(0)$ from the last few points of the inward integration grid. This is important because a lower-order boundary derivative would spoil the fourth-order behavior otherwise expected from the Numerov recurrence in the harmonic-oscillator study.

3 Results

Each potential is used to validate, test, or demonstrate different aspects of the numerical method, ranging from analytic benchmarks to nontrivial quantum behavior.

3.1 In nite square well

The infinite square well serves as a benchmark problem with known analytic spectrum:

$$E_n = \frac{(n+1)^2 \ ^2}{8a^2} \tag{3}$$

The potential is:

$$V(x) = \begin{cases} 0 & ; |x| \ a \\ 1 & ; |x| > a \end{cases} \qquad \text{(implemented numerically with a large finite } V_0) \tag{4}$$

In the production calculation, the well half-width is $a = 1$, the outward-shooting half-domain uses $x_{max} = a = 1$, the Numerov grid uses $n_{grid} = 2500$ points, and the numerical wall height is set to $V_0 = 10^6$. The grid-convergence study for this case uses $n_{grid} \ 2 \{50, 80, 120, 180\}$ at fixed $x_{max} = 1$.

The numerical eigenvalues agree with the analytic values to essentially machine precision, as shown in Fig. 1. In the saved run the first four relative errors are $4{:}9 \ \ 10^{-11}$, $1{:}3 \ \ 10^{-10}$,

2:6 10 ^{11} , and 7:0 10 ^{12} . This is exactly the behavior expected for a benchmark problem of this type: the energies follow the analytic law $E_n / (n+1)^2$, and the numerical points are visually indistinguishable from the exact curve on the scale of the plot.

The probability densities for the first four states (Fig. 2) also have the expected structure. The ground-state density is maximal at the center, while the excited states show the correct sequence of interior nodes inherited from the alternating even/odd parity eigenfunctions. All densities vanish at $x = a$, consistent with the hard-wall boundary condition.

The shooting method is validated by plotting the boundary mismatch functions for both even and odd sectors (Fig. 3). The even-sector scan has zero crossings near E 1:234 and E 11:103, while the odd-sector scan crosses zero near E 4:935 and E 19:739. These are precisely the first four physical eigenvalues, so the parity separation and bracketing logic are behaving as intended rather than picking spurious roots.

The convergence of the energy error as a function of grid spacing h (Fig. 5) is also as expected for Numerov. The fitted exponents from the saved convergence study are p = 3:94, 4:00, 3:98, and 4:00 for the first four states, which is strong evidence that the recurrence, startup values, and boundary treatment are all consistent with fourth-order accuracy. The higher states sit at larger absolute error for a fixed grid, which is also expected because their shorter wavelengths are harder to resolve.

Finally, the wavefunctions for the first four states are plotted together with their corresponding energies and the confining potential (Fig. 6). The alternating parity, increasing node count, and quadratic growth of the energy levels all match the analytic infinite-well picture.

3.2 Harmonic oscillator

3.2.1 Numerov method on unbounded domains

For the harmonic oscillator with $V(x) = \frac{1}{2}! \, ^2x^2$, the exact spectrum is:

$$E_n = ! \quad n + \tfrac{1}{2} \; ; \qquad n = 0; 1; 2; : : : \tag{5}$$

This provides a second analytic benchmark, but on an unbounded domain. Numerically, the domain is truncated to $[x_{max}; x_{max}]$, and decaying boundary conditions are imposed. In the production Numerov run, the parameters are $! = 1$, $x_{max} = 8$, and $n_{grid} = 2500$. The grid-convergence study uses $n_{grid} \; 2 \, f500; 800; 1200; 1800; 2500g$ at fixed $x_{max} = 8$, while the box-size study uses $x_{max} \; 2 \, f4; 5; 6; 7; 8; 10g$ at approximately fixed spacing.

The numerical eigenvalues agree with the exact linear spectrum to extremely high precision, as shown in Fig. 7. In the saved run the first four relative errors are 6:2 10 ^{13} , 2:3 10 ^{12} , 6:6 10 ^{12} , and 7:5 10 ^{12} . This is exactly what should happen for the harmonic oscillator benchmark: the inward-shooting formulation suppresses contamination by the exponentially growing forbidden-region solution and cleanly recovers the exact ladder $E_n = !(n + \frac{1}{2})$.

The probability densities (Fig. 8) show the expected structure: the ground state is centered and Gaussian-like, while the excited states alternate in parity and gain one additional node at each step. The support also broadens with increasing energy, consistent with the classical turning points moving outward. This confirms both physical correctness and stability of the integration on the truncated domain.

The shooting mismatch functions (Fig. 9) exhibit clear zero crossings at the expected energies. In the even sector the roots lie near E 0:5 and E 2:5, while in the odd sector they lie near

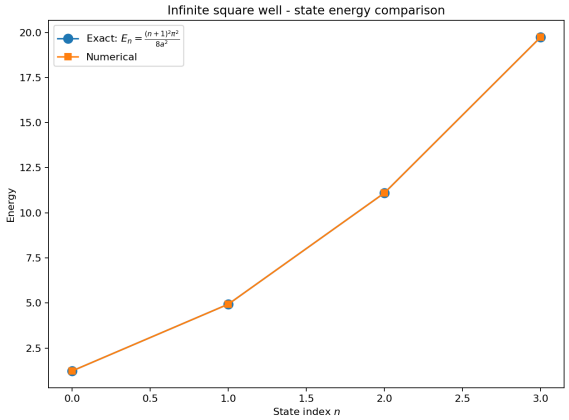


Figure 1: Comparison between numerical and exact energy eigenvalues for the infinite square well. The agreement is excellent for the first four states.

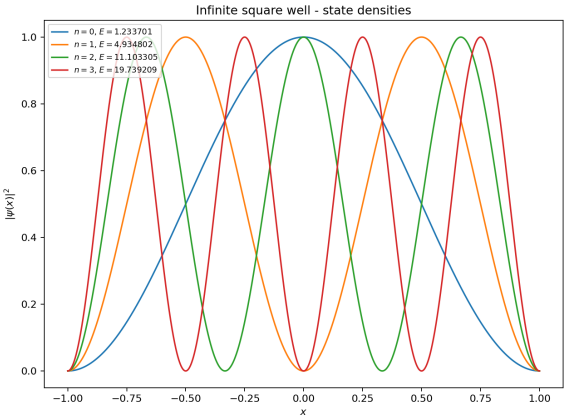
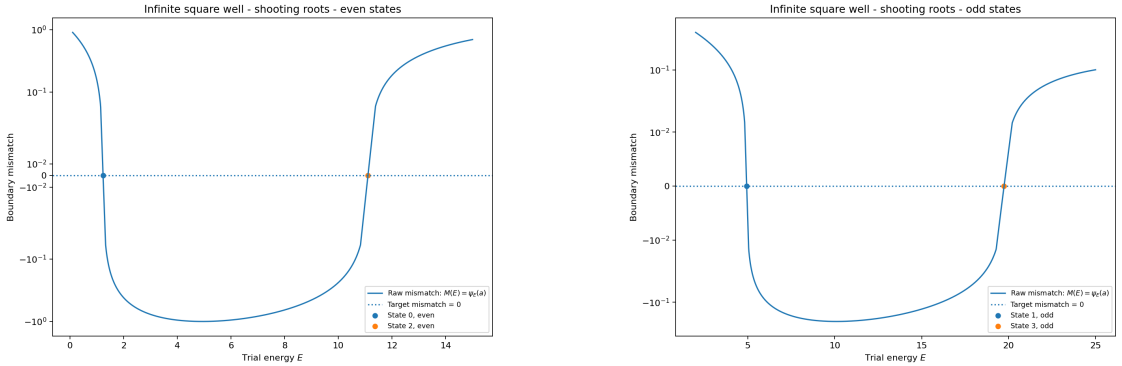
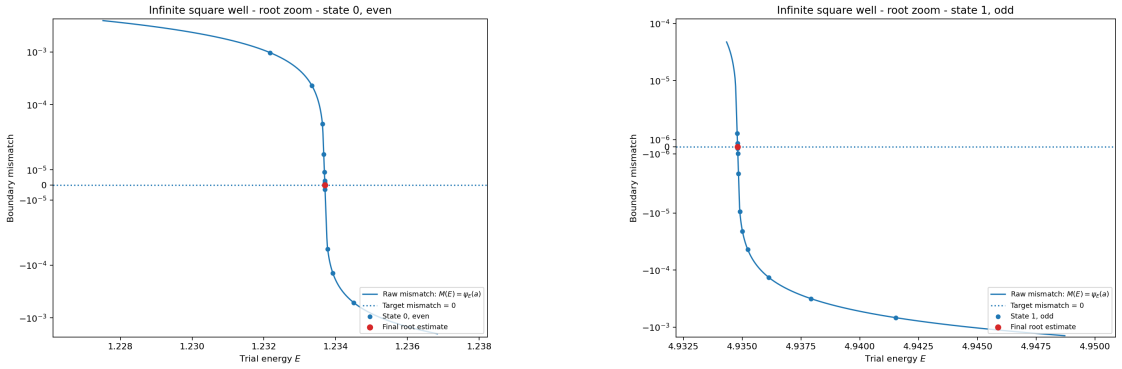


Figure 2: Probability densities $|\psi(x)|^2$ for the first four eigenstates of the infinite square well. The correct number of nodes and boundary behavior are clearly reproduced.



(a) (b)

Figure 3: Raw boundary mismatch functions for even (a) and odd (b) states. The zero crossings correspond to the physical eigenvalues found by the shooting method.



(a) (b)

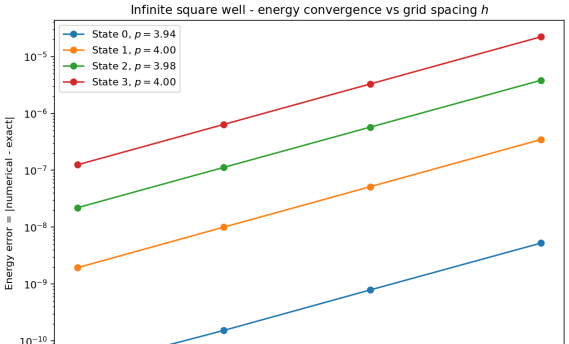
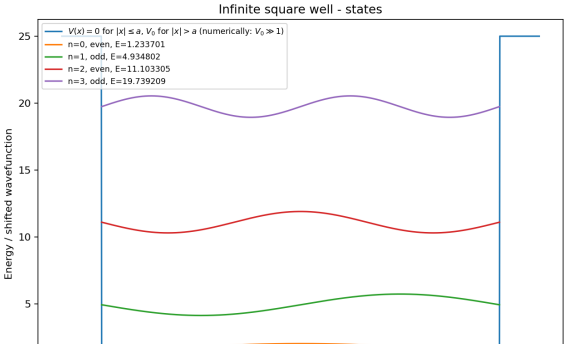



Figure 4: Zoomed root brackets for the lowest even (a) and odd (b) infinite-well states. These panels show the raw mismatch only inside one initial sign-changing bracket, together with the recorded bisection iterates.